

JavaScript Tasks – Type Casting & Flow Control

Task 1 – Predict the Output

1.

```
console.log("10" + 5);
```

Output: 105

Datatype: string

2.

```
console.log(10 + true);
```

Output: 11

Datatype: number

3.

```
console.log(false + null);
```

Output: 0

Datatype: number

4.

```
console.log("Hello" + undefined);
```

Output: Helloundefined

Datatype: string

5.

```
console.log([1,2] + 5);
```

Output: 1,25

Datatype: string

Task 2 – Implicit Type Casting

String + Number

```
let ex1 = "20" + 10;  
console.log(ex1, typeof ex1);
```

Output: 2010 string

Boolean + Number

```
let ex2 = true + 10;  
console.log(ex2, typeof ex2);
```

Output: 11 number

Array + String

```
let ex3 = [1, 2] + "hello";  
console.log(ex3, typeof ex3);
```

Output: 1,2hello string

Object + Number

```
let ex4 = {} + 10;  
console.log(ex4, typeof ex4);
```

Output: [object Object]10 string

Null + Number

```
let ex5 = null + 10;  
console.log(ex5, typeof ex5);
```

Output: 10 number

Task 3 – Explicit Type Casting

```
console.log(Number("500"));
```

Output: 500

```
console.log(Number(true));
```

Output: 1

```
console.log(Number(false));
```

Output: 0

```
console.log(Number(null));
```

Output: 0

```
console.log(Number("abc"));
```

Output: NaN

```
console.log(Number([100]));
```

Output: 100

Task 4 – Boolean Constructor

```
console.log(Boolean(""));
```

Output: false

```
console.log(Boolean("javascript"));
```

Output: true

```
console.log(Boolean(0));
```

Output: false

```
console.log(Boolean(1));
```

Output: true

```
console.log(Boolean(null));
```

Output: false

```
console.log(Boolean(undefined));
```

Output: false

```
console.log(Boolean([]));
```

Output: true

```
console.log(Boolean({}));
```

Output: true

Task 5 – Student Pass or Fail

```
let mark = 45;  
if (mark > 35) {  
  console.log("Pass");  
} else {  
  console.log("Fail");  
}
```

Output: Pass

Task 6 – Voting Eligibility

```
let age = 20;

if (age >= 18) {
  console.log("Eligible");
} else {
  console.log("Not Eligible");
}
```

Output: Eligible

Task 7 – Greatest Number

```
let a = 50;
let b = 80;
let c = 30;

if (a > b && a > c) {
  console.log("A is greatest");
} else if (b > a && b > c) {
  console.log("B is greatest");
} else {
  console.log("C is greatest");
}
```

Output: B is greatest

Task 8 – Traffic Light System

```
let light = "red";

switch (light) {
  case "red":
    console.log("Stop");
    break;

  case "yellow":
    console.log("Ready");
    break;

  case "green":
    console.log("Go");
    break;

  default:
    console.log("Invalid light");
}
```

Output: Stop

Task 9 – Login System

```
let username = "admin";  
let password = "1234";  
  
if (username === "admin") {  
  if (password === "1234") {  
    console.log("Login Success");  
  } else {  
    console.log("Invalid Login");  
  }  
} else {  
  console.log("Invalid Login");  
}
```

Output: Login Success

Task 10 – Session Finder

```
let hour = 14;  
  
if (hour >= 1 && hour <= 12) {  
  console.log("Morning");  
} else if (hour >= 13 && hour <= 15) {  
  console.log("Afternoon");  
} else if (hour >= 16 && hour <= 19) {  
  console.log("Evening");  
} else if (hour >= 20 && hour <= 24) {  
  console.log("Night");  
}
```

```
} else {  
  console.log("Invalid Hour");  
}
```

Output: Afternoon

Bonus Challenge

1.

```
console.log(true + true);
```

Output: 2

2.

```
console.log("5" - 2);
```

Output: 3

3.

```
console.log("5" + 2);
```

Output: 52

4.

```
console.log(null + 1);
```

Output: 1

5.

```
console.log(undefined + 1);
```

Output: NaN

6.

```
console.log(Boolean(" "));
```

Output: true

7.

```
console.log(Number(true));
```

Output: 1